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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:            0.0
Usage of /:             5.1% of 48.91GB
Memory usage:          5%
Swap usage:            0%
Processes:             139
Users logged in:       1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Student ID: 202211691

Course/Year/Section: BSIT 3B

Date Submitted: 2/09/2023

Instructor: Johnerus P.Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	ubuntu
Virtualization Software (VirtualBox/VMware)	Oracle Virtual box
Ubuntu Server Version	Ubuntu-26.04 live server
VM Name	rdserver
RAM Allocated	8000
Storage Allocated	100
Network Mode	Bridged adapter
SSH Client Used	Windows PowerShell

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	_____8000__ MB
Hard Disk	____100____ GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION = ip a

Commands Executed: ip a

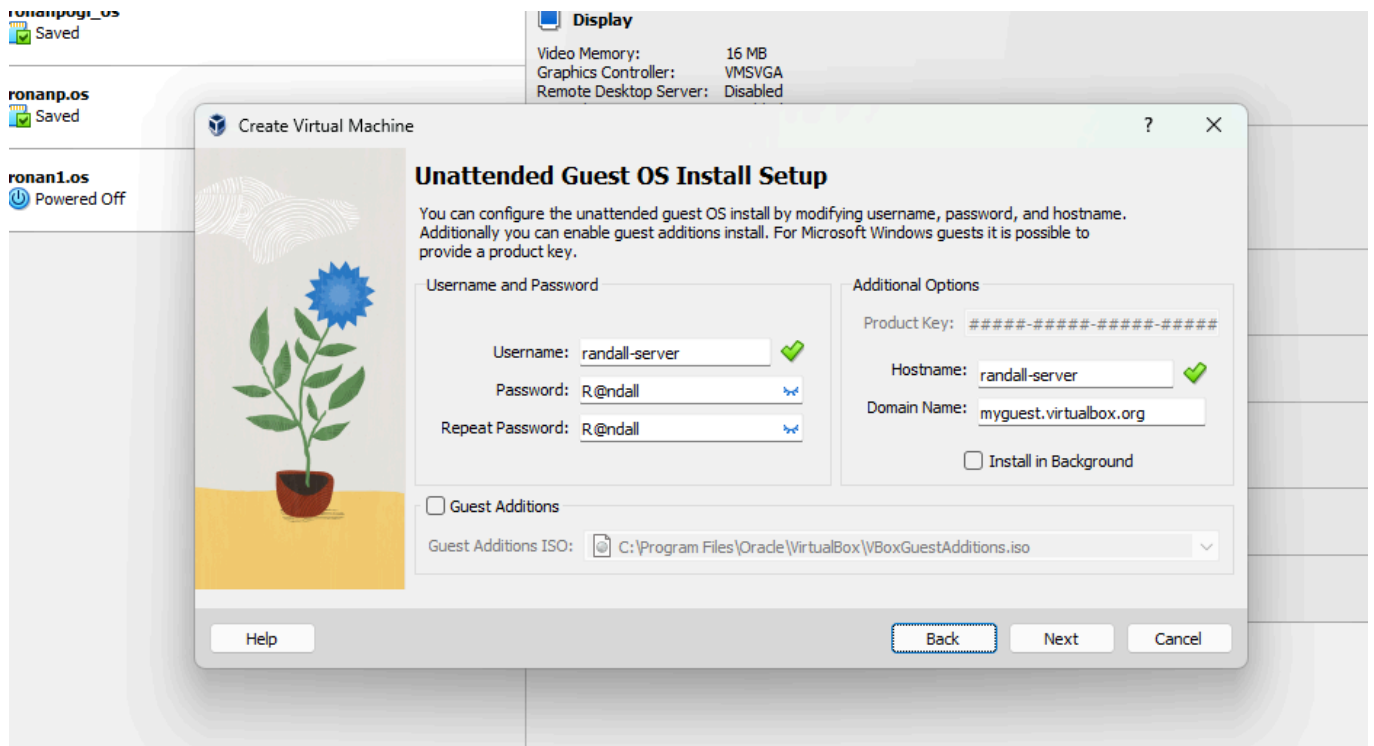
Server IP Address: 192.168.56.101

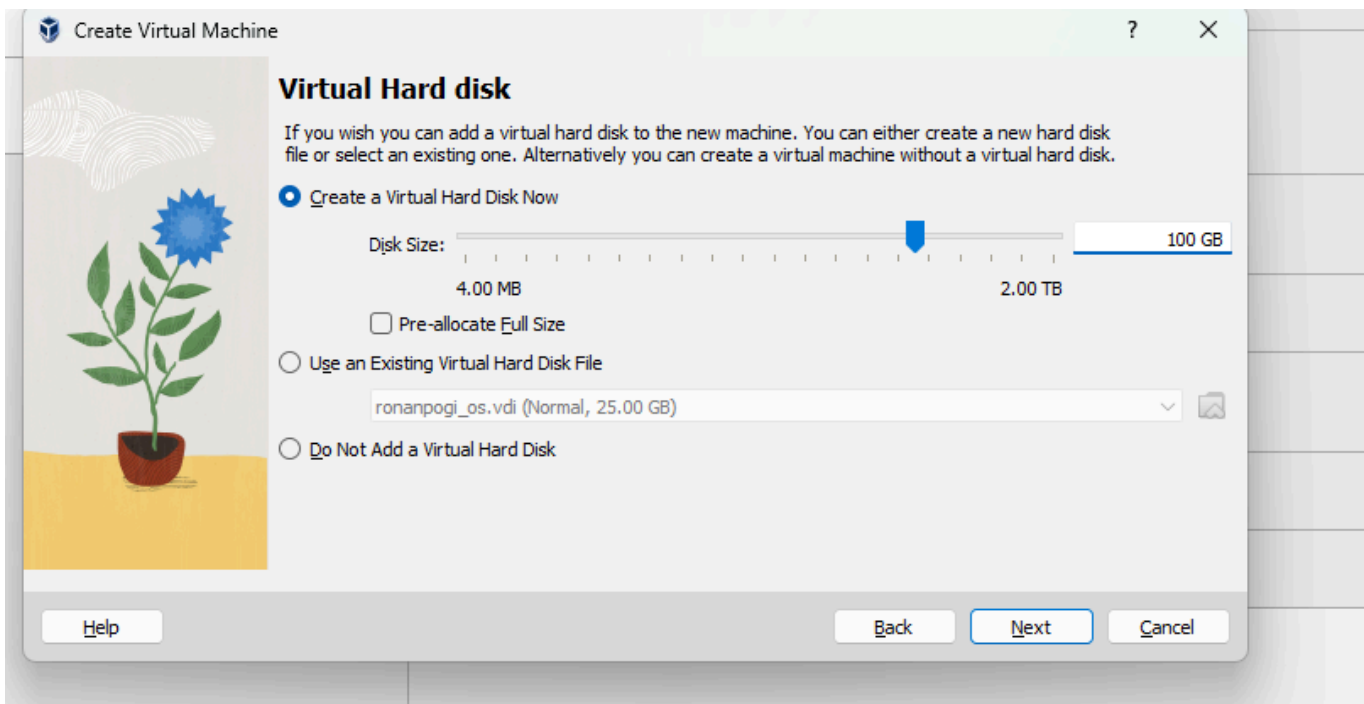
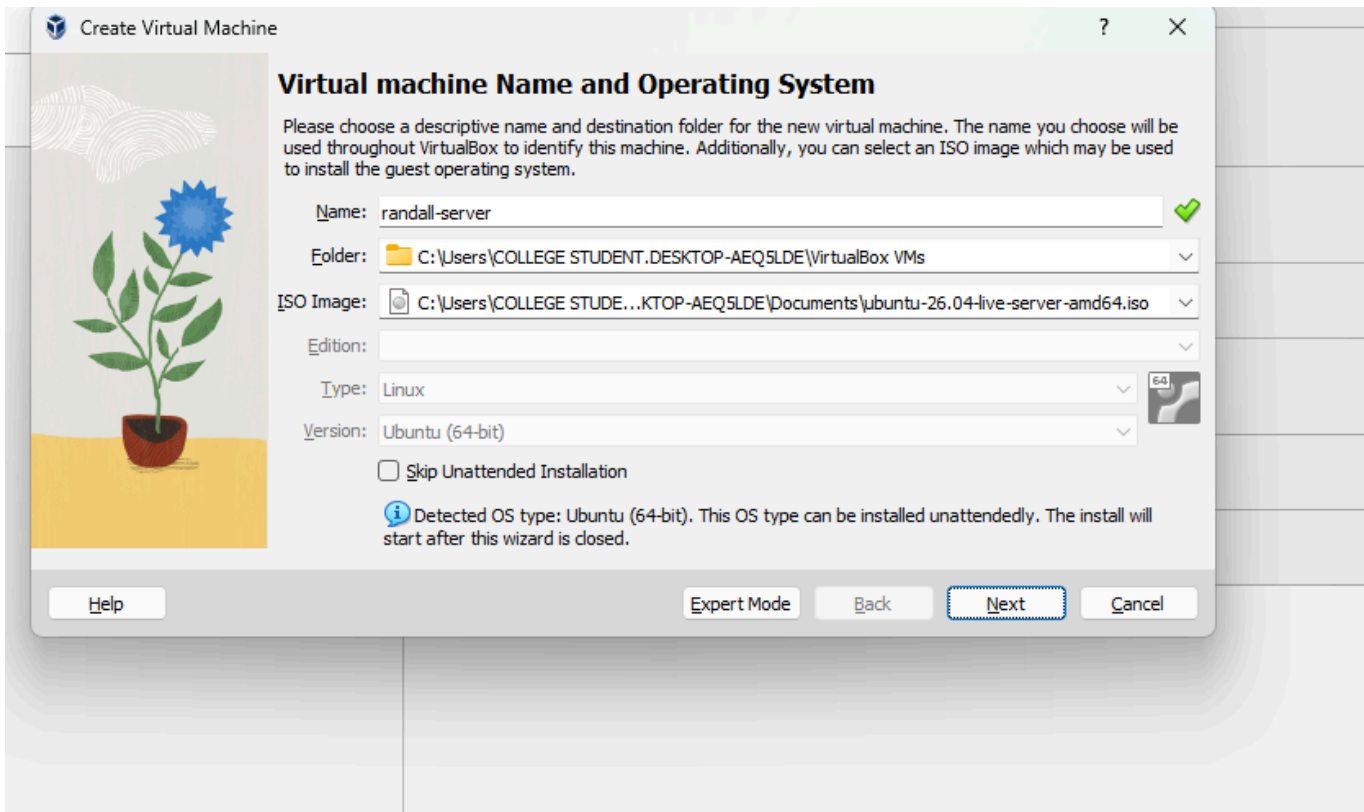
Network Interface: enp0s3

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.56.101
Username	randall
SSH Port	22







Create Virtual Machine



Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.



Machine Name and OS Type

Machine Name	randall-server
Machine Folder	C:/Users/COLLEGE STUDENT.DESKTOP-AEQ5LDE/VirtualBox VMs/randall...
ISO Image	C:/Users/COLLEGE STUDENT.DESKTOP-AEQ5LDE/Documents/ubuntu-2...
Guest OS Type	Ubuntu (64-bit)
Skip Unattended Install	false



Unattended Install

Username	randall-server
Product Key	false
Hostname/Domain Name	randall-server.myquest.virtualbox.org
Install in Background	false
Install Guest Additions	false



Hardware

Base Memory	800
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[Help](#)[Back](#)[Finish](#)[Cancel](#)

randall-server [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!

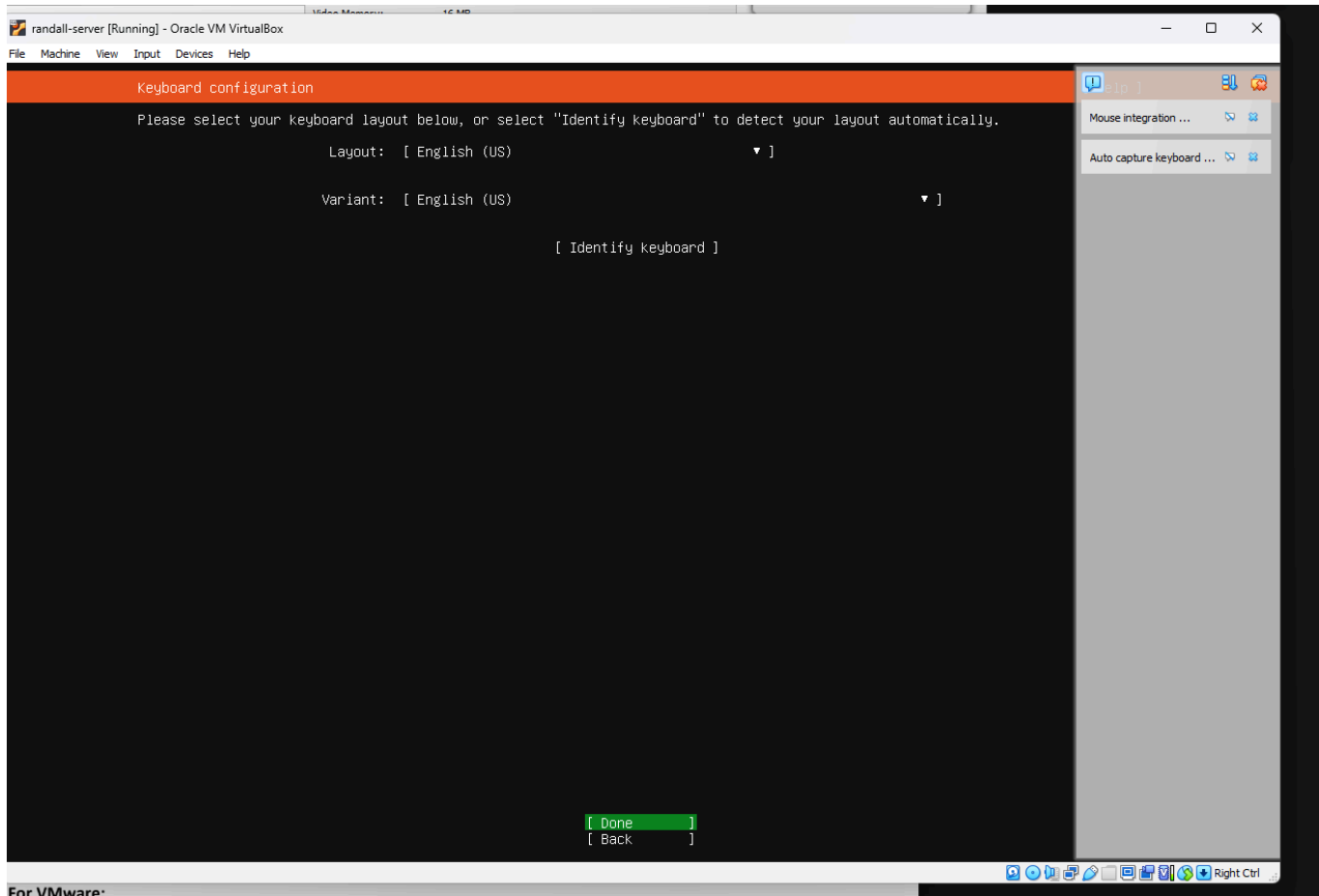
Use UP, DOWN and ENTER keys to select your language.

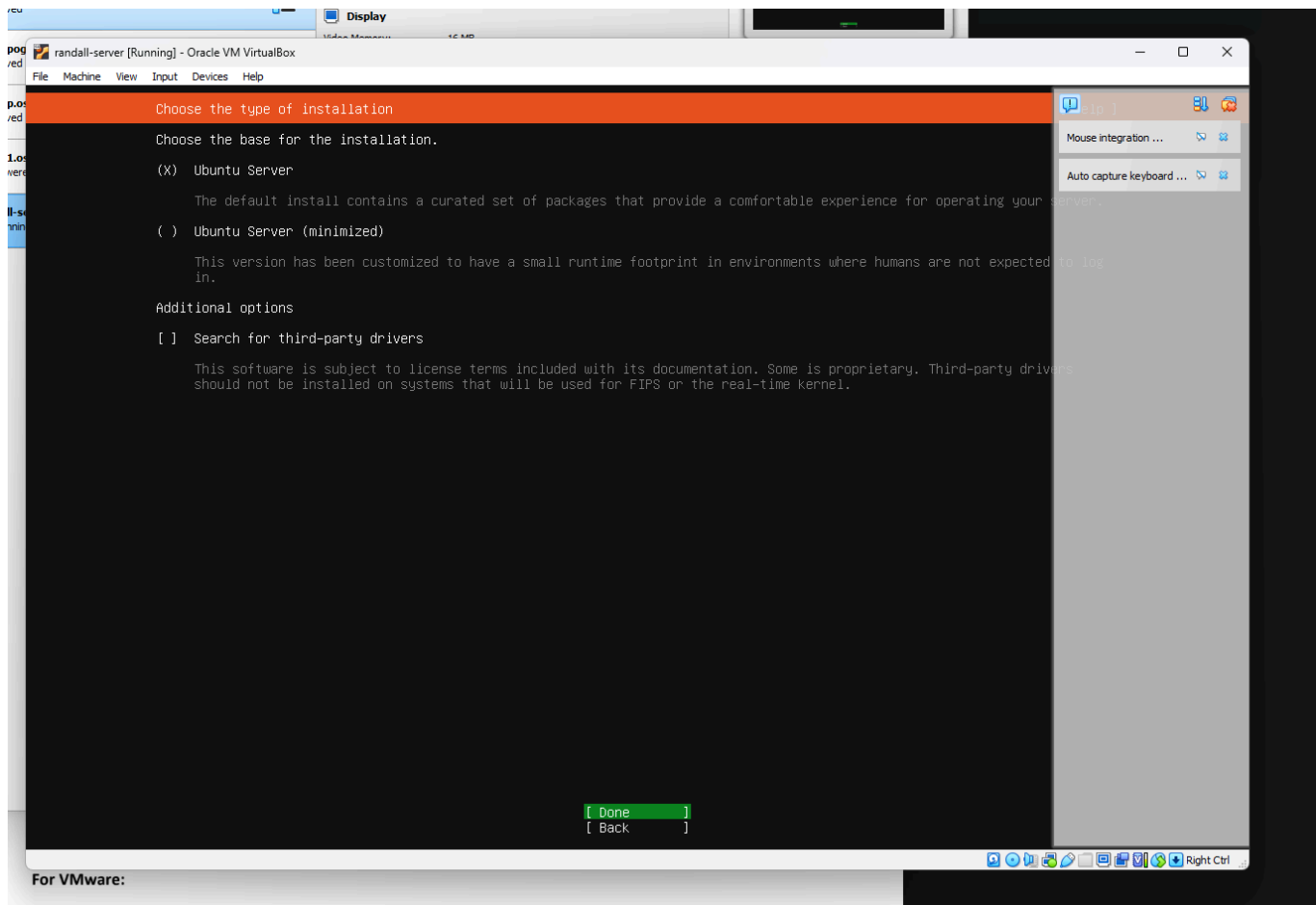
```
[ Asturianu
[ Bahasa Indonesia
[ Català
[ Deutsch
[ English
[ English (UK)
[ Español
[ Français
[ Galego
[ Hrvatski
[ Latviski
[ Lietuviškai
[ Magyar
[ Nederlands
[ Norsk bokmål
[ Occitan
[ Polski
[ Português
[ Suomi
[ Svenska
[ Čeština
[ Ελληνικά
[ Беларуская
[ Русский
[ Српски
[ Українська
```

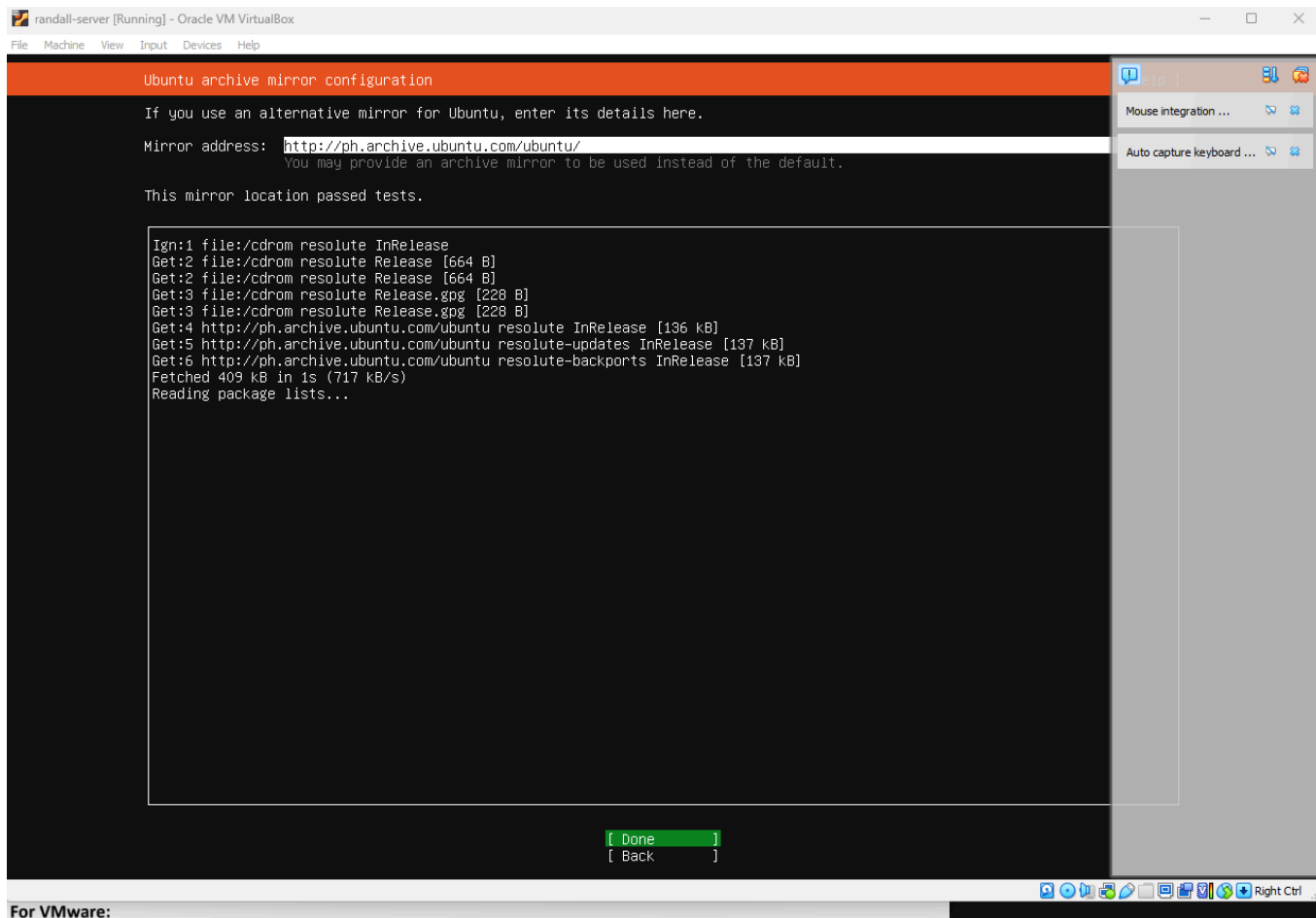


Mouse integration ...

Auto capture keyboard ...







randall@randallserver: ~ — ssh randall@192.168.56.1...

randall@randall-VirtualBox:~\$ ssh randall@192.168.56.101

randall@192.168.56.101's password:

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

- * Documentation: <https://docs.ubuntu.com>
- * Management: <https://landscape.canonical.com>
- * Support: <https://ubuntu.com/pro>

System information as of Wed Sep 2 07:49:23 AM UTC 2026

System load: 0.0	Memory usage: 3%	Processes: 143
Usage of /: 15.0% of 47.93GB	Swap usage: 0%	Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.

To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

randall@randallserver:~\$

```
rdserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Sep 02 06:44:52 randallserver systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 02 06:44:52 randallserver sshd[5566]: Server listening on 0.0.0.0 port 22.
Sep 02 06:44:52 randallserver sshd[5566]: Server listening on :: port 22.
Sep 02 06:44:52 randallserver systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
randall@randallserver:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
randall@randallserver:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55
randall@randallserver:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
randall@randallserver:~$ sudo systemctl start ssh
randall@randallserver:~$
  apt      install      ssh      status      systemctl      -y
enable    openssh-server  start    sudo         update
randall@randallserver:~$
  apt      install      ssh      status      systemctl      -y
enable    openssh-server  start    sudo         update
randall@randallserver:~$
  apt      install      ssh      status      systemctl      -y
enable    openssh-server  start    sudo         update
randall@randallserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:dc:3e:f5 brd ff:ff:ff:ff:ff:ff
    altname enx080027dc3ef5
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85987sec preferred_lft 85987sec
    inet6 fe80::a00:27ff:fedc:3ef5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
randall@randallserver:~$
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randall@randallserver: ~ — ssh randall@192.168.56.1...  
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randall@randall-VirtualBox:~$ ssh randall@192.168.56.101  
randall@192.168.56.101's password:  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Wed Sep  2 07:49:23 AM UTC 2026  
  
System load: 0.0           Memory usage: 3%    Processes:      143  
Usage of /:  15.0% of 47.93GB Swap usage:   0%    Users logged in: 0  
  
Expanded Security Maintenance for Applications is not enabled.  
  
53 updates can be applied immediately.  
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randall@randallserver: ~ — ssh randall@192.168.56.1...  
~  
randall@randall-VirtualBox:~$ ssh randall@192.168.56.101  
randall@192.168.56.101's password:  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)  
  
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Usage of /:  15.0% of 47.93GB  Swap usage:  0%    Users logged in: 0  
  
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```